

ELEC 221 Tutorial 1 – Solutions

Signals, Time Transformations, and System Properties

Monday, 14 September 2026

Overview

- 1 CT vs. DT signals
- 2 Transformations of the independent variable
- 3 System properties

1.1 – Continuous-time or discrete-time?

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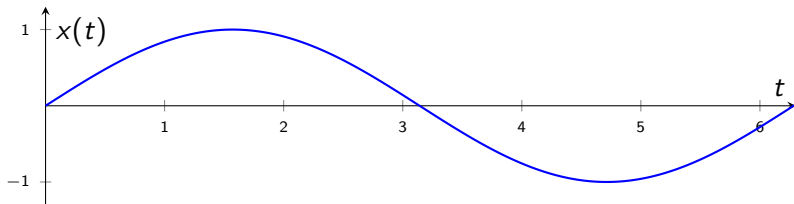
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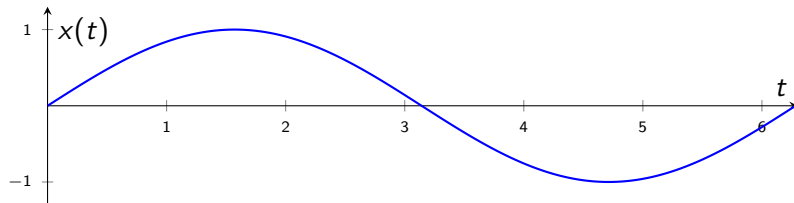
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- (d) **Heart rate:** Individual heartbeats are discrete events, and “rate” is usually computed from beat-to-beat intervals $\Rightarrow$ *ambiguous*: the underlying electrical signal is CT, but the derived “rate” is naturally a DT sequence (one value per beat).

1.2 – Discretizing $x(t) = \sin(t)$



(a) $t = 0, 1, \dots, 6 \Rightarrow x[n] = \sin(n)$: $\{0, .84, .91, .14, -.76, -.96, -.28\}$

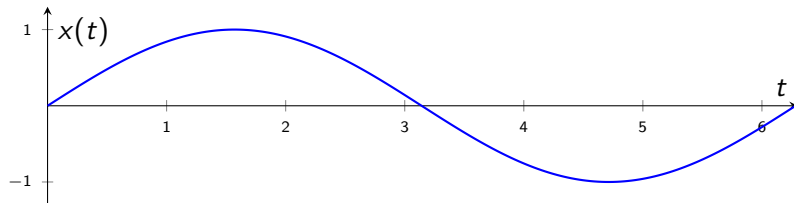
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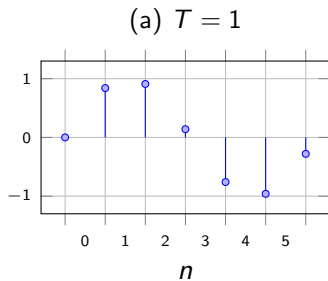
(b) $t = 0, .5, \dots, 6 \Rightarrow x[n] = \sin(0.5n)$, $n = 0, \dots, 12$ (finer grid)

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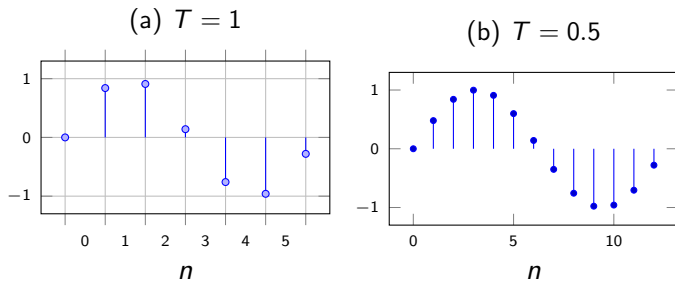
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- (b) $t = 0, .5, \dots, 6 \Rightarrow x[n] = \sin(0.5n), n = 0, \dots, 12$ (finer grid)
- (c) $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi \Rightarrow x[n] = \sin(n\pi/2) = \{0, 1, 0, -1, 0\}$

1.2 – Qualitative comparison



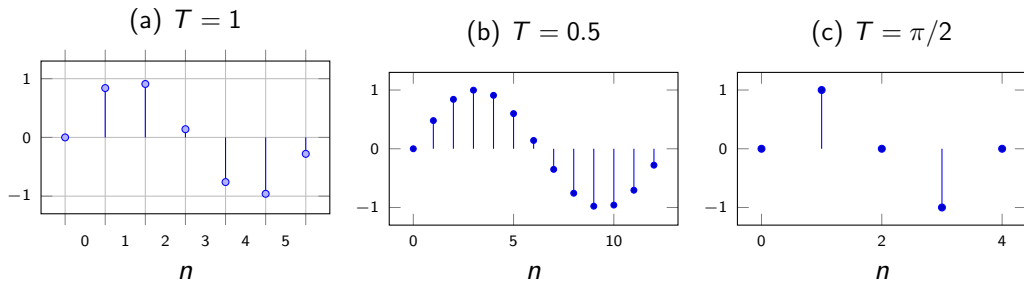
- (a) Coarse sampling

1.2 – Qualitative comparison



- (a) Coarse sampling
- (b) Fine sampling

1.2 – Qualitative comparison



- (a) Coarse sampling
- (b) Fine sampling
- (c) Only 5 points, landing exactly on zero-crossings/peaks

Method: “shift, scale, reverse”

To obtain $x(\alpha t + \beta)$ from $x(t)$:

$$x(t) \xrightarrow{\text{shift by } \beta} x_1(t) = x(t + \beta) \xrightarrow{\text{scale by } |\alpha|} x_2(t) = x_1(|\alpha|t) \xrightarrow{\text{reverse if } \alpha < 0} x_3(t) = x_2(-t)$$

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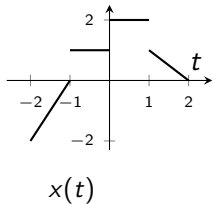
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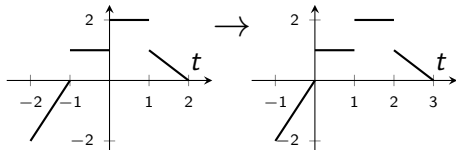
Same idea in discrete time for $x[an + b]$:

$$x[n] \xrightarrow{\text{shift by } b} x_1[n] = x[n + b] \xrightarrow{\text{scale by } a} x_1[an]$$

$x(-0.5t - 1)$: shift, scale, reverse



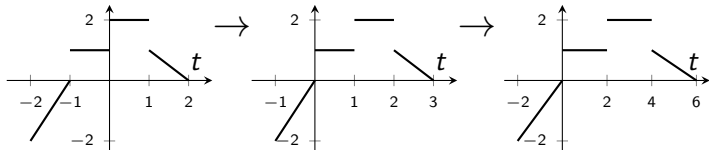
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$$x(t) \quad x_1(t) = x(t - 1)$$

shift by $\beta = -1$

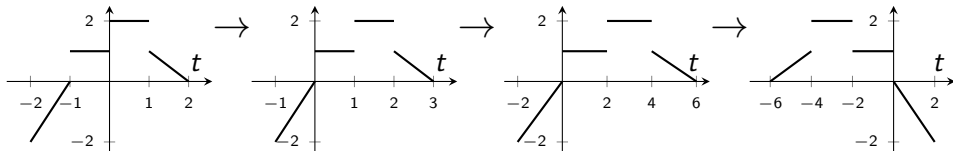
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$$x(t) \quad x_1(t) = x(t - 1) \quad x_2(t) = x(0.5t - 1)$$

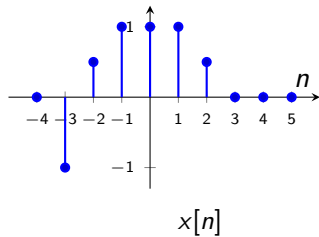
shift by $\beta = -1$ scale by $|\alpha| = 0.5$

$x(-0.5t - 1)$: shift, scale, reverse

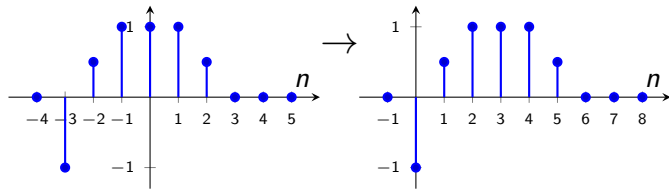


$$\begin{array}{lll}
 x(t) & x_1(t) = x(t-1) & x_2(t) = x(0.5t-1) \\
 & \text{shift by } \beta = -1 & \text{scale by } |\alpha| = 0.5 \\
 & & x_3(t) = x(-0.5t-1) = y(t) \\
 & & \text{reverse } (\alpha < 0)
 \end{array}$$

$x[2n - 3]$: shift then compress



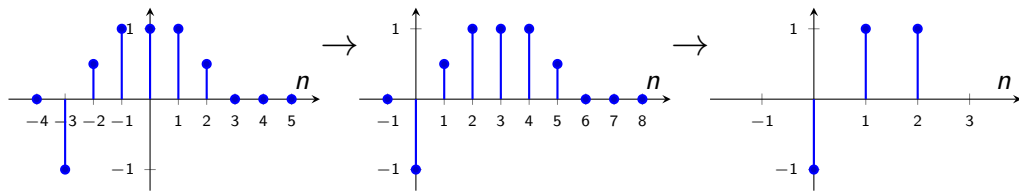
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$$x_1[n] = x[n-3]$$

shift by 3

$x[2n - 3]$: shift then compress



$$x[n] \quad x_1[n] = x[n - 3]$$

shift by 3

$$y[n] = x_1[2n] = x[2n - 3]$$

compress by 2 (keep even-index samples)

Compression *discards* samples – only odd indices of x survive here.

3.1 – Memory and causality

(a) $y(t) = \cos(3t)x(t)$:

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- (c) $y[n] = (n+1)(x[n] - x[n-1])$:
uses present *and* one past sample $\Rightarrow$ **not memoryless** (has memory), but only present/past $\Rightarrow$ **causal**.

3.2 – Invertibility

Recall: invertible $\iff$ distinct input *signals* give distinct output signals.

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(c) $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$:

invertible (zero-adding / upsampling).

3.3 – Stability (BIBO)

Assume $|x(\cdot)| \leq B < \infty$.

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(d) $y[n] = n^{10}x[n]$:

even for constant bounded $x[n] = B$, $y[n] = n^{10}B \rightarrow \infty$ as $n \rightarrow \infty \Rightarrow$ **not stable**.